



CS683: Advanced Computer Architecture

Lecture 11: How to do Computer Architecture Research(projects)

<https://www.cse.iitb.ac.in/~biswa/courses/>

<https://www.cse.iitb.ac.in/~biswa/>



Research/Project

Read Top conference papers only. Period!

ISCA

MICRO

HPCA

ASPLOS

PACT, IPDPS, ICS

ISPASS

Top venues in the field of Computer Architecture

For projects

- Conferences not the in the previous slides 😊
- So for your projects, browse recent ISCA, MICRO, and HPCA and ASPLOS (papers are architecture+systems)

<https://iscaconf.org/>

<https://microarch.org/>

<https://www.hpca-conf.org/>

<https://www.asplos-conference.org/>

Why top conferences?

- Imagine eating a low-quality Pizza/Dosa/Biryani/Vada-pav 😊
- Same applies to these conferences. These conferences push the state-of-the-art (Intellectual food)
- Look for insights instead of numbers

Computer Architecture Research in India

(Top two conferences, goal is to show groups and not ranks)

Rank institutions in by publications from to

All Areas [off | on]

AI [off | on]

- ▶ Artificial intelligence ☐
- ▶ Computer vision ☐
- ▶ Machine learning ☐
- ▶ Natural language processing ☐
- ▶ The Web & information retrieval ☐

Systems [off | on]

- ▼ Computer architecture ☒
- ACM SIGARCH
- ASPLOS ☐
- ISCA ☒
- MICRO ☒
- HPCA ☐

+Intel PARL lab

#	Institution	Count	Faculty
1	▼ IIT Bombay  	3.0	1
	<u>Faculty</u>	<u>#..PubsAdj..#</u>	
	Biswabandan Panda ARCH   	4	2.0
2	▼ IISc Bangalore  	2.4	3
	<u>Faculty</u>	<u>#..PubsAdj..#</u>	
	Arkaprava Basu ARCH   	3	1.0
	Uday Bondhugula HPC,ARCH 	1	0.2
	 		
	R. Govindarajan ARCH   	1	0.2
3	▼ IIT Kanpur  	2.0	1
	<u>Faculty</u>	<u>#..Pubs Adj..#</u>	
	Mainak Chaudhuri   	1	1.0



Goal of research?

To find insights

Keywords:

Why?

How?

Why not?

..

A hand holding a compass over a desert landscape with a winding road and a body of water.

Before that some basics to start with

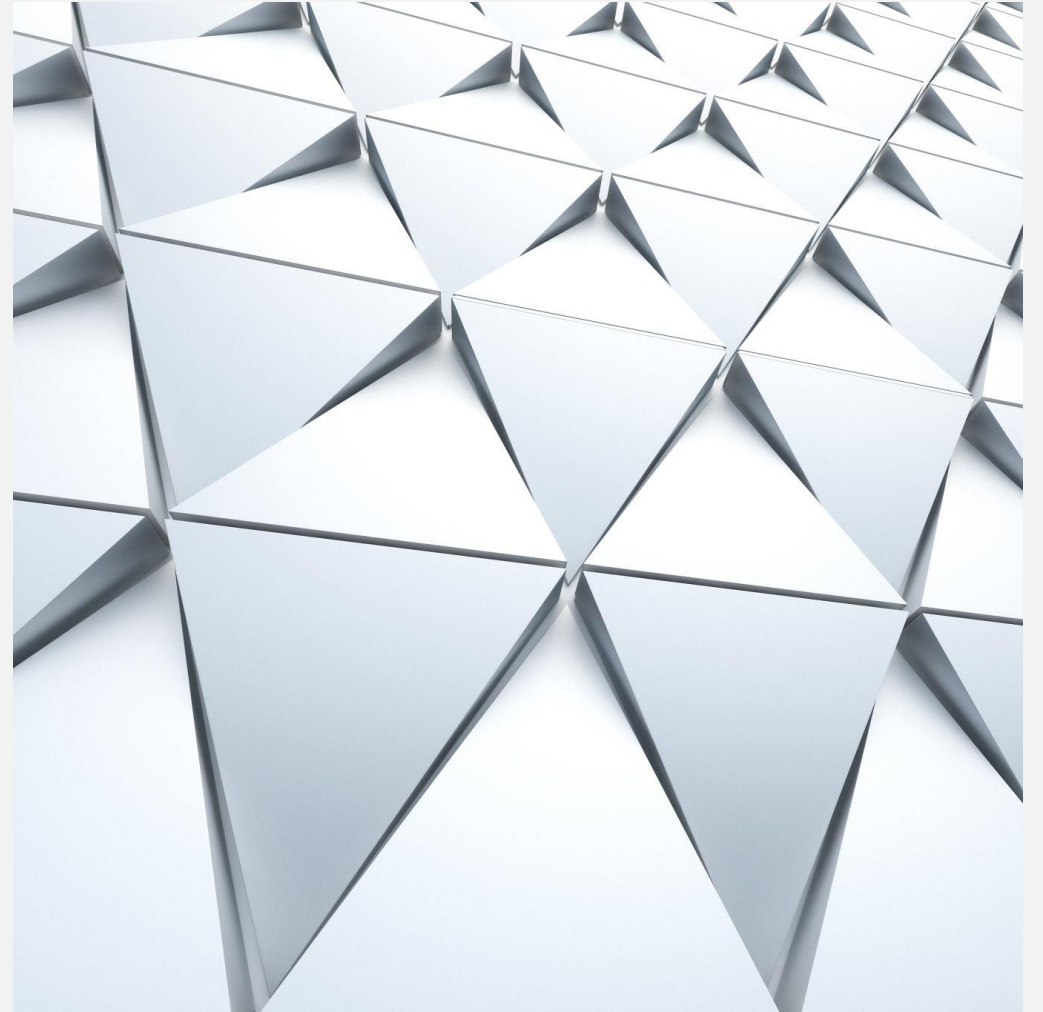
Know the Problem domain

Know the metrics of interest

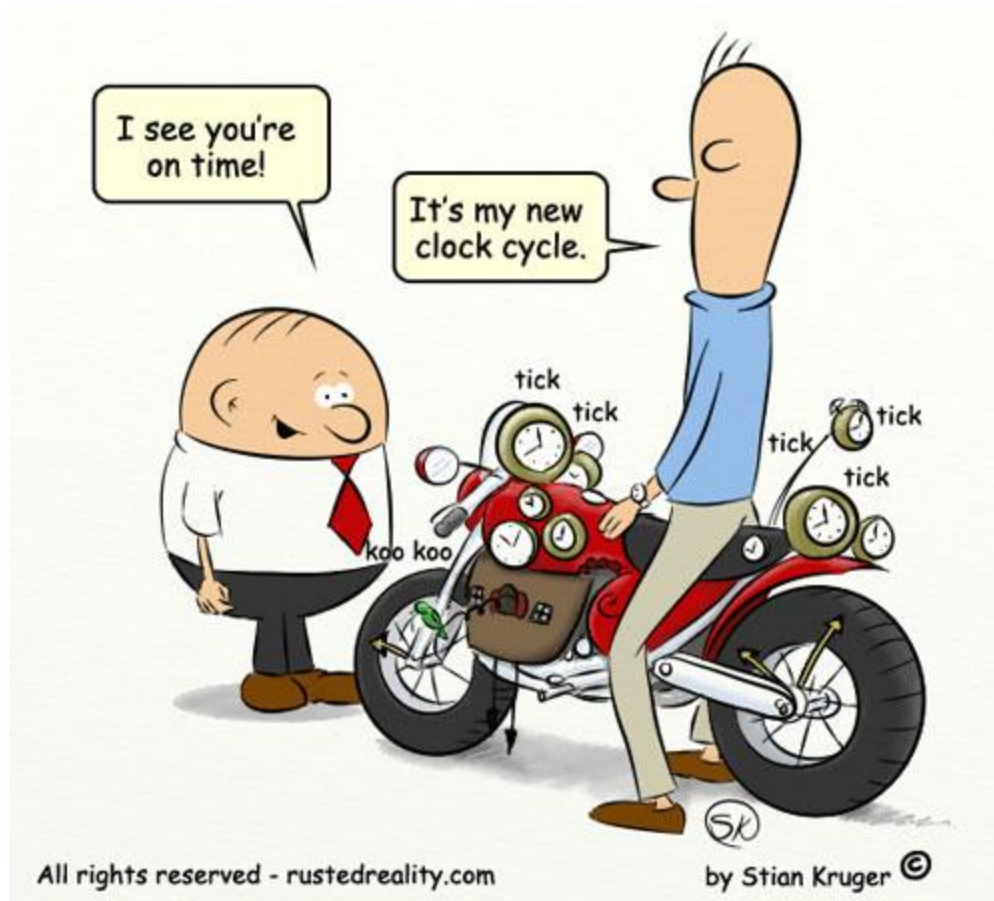
Performance: IPC, Execution time

Power: Performance per watt

Security: Know your threat model
else you will be fooled 😊



Performance



Benchmarks

Metrics

Simulators

Latency and bandwidth



Amdahl's law

- Common case
- Uncommon case

Evaluation

- To compare Processor A with Processor B by running programs
- How many programs? 
- The programs that you care. 
- What if I want to build a new one (processor, caches, DRAM) ? 

World of Benchmarks

- SPEC CPU 2017 (<https://www.spec.org/cpu2017/>)

The **SPEC CPU® 2017** benchmark package contains SPEC's **next-generation, industry-standardized, CPU intensive** suites for measuring and comparing **compute intensive performance**, stressing a system's processor, memory subsystem and compiler.

SPECspeed: used for comparing time for a computer to complete single tasks

SPECrate: measure the throughput or work per unit of time.

World of Benchmarks

CloudSuite (<https://www.cloudsuite.ch/>)

CloudSuite is a benchmark suite for **cloud services**. The benchmarks are based on real-world software stacks and represent real-world setups.

PARSEC (<https://parsec.cs.princeton.edu/>)

Benchmark suite composed of **multithreaded** programs. The suite focuses on emerging workloads and was designed to be representative of next-generation shared-memory programs for chip-multiprocessors.

World of Benchmarks

MobileBench

(<https://mobilebench.engineering.asu.edu/>)

comprising a selection of representative **smart phone applications**.

Many more application domain specific: Graph processing, ML perf,

Pitfalls of Benchmarks

Benchmark not representative of all

Your workload is I/O bound → SPEC CPU is useless

Benchmark is too old

Need to be periodically refreshed

Non-benchmarks

- Application kernels: A small code fragment or part of the program
- Synthetic benchmark : Not part of any real program!!
- Micro-benchmark
- *OK! So, I will create a chip and then evaluate these benchmarks*

World of Simulators

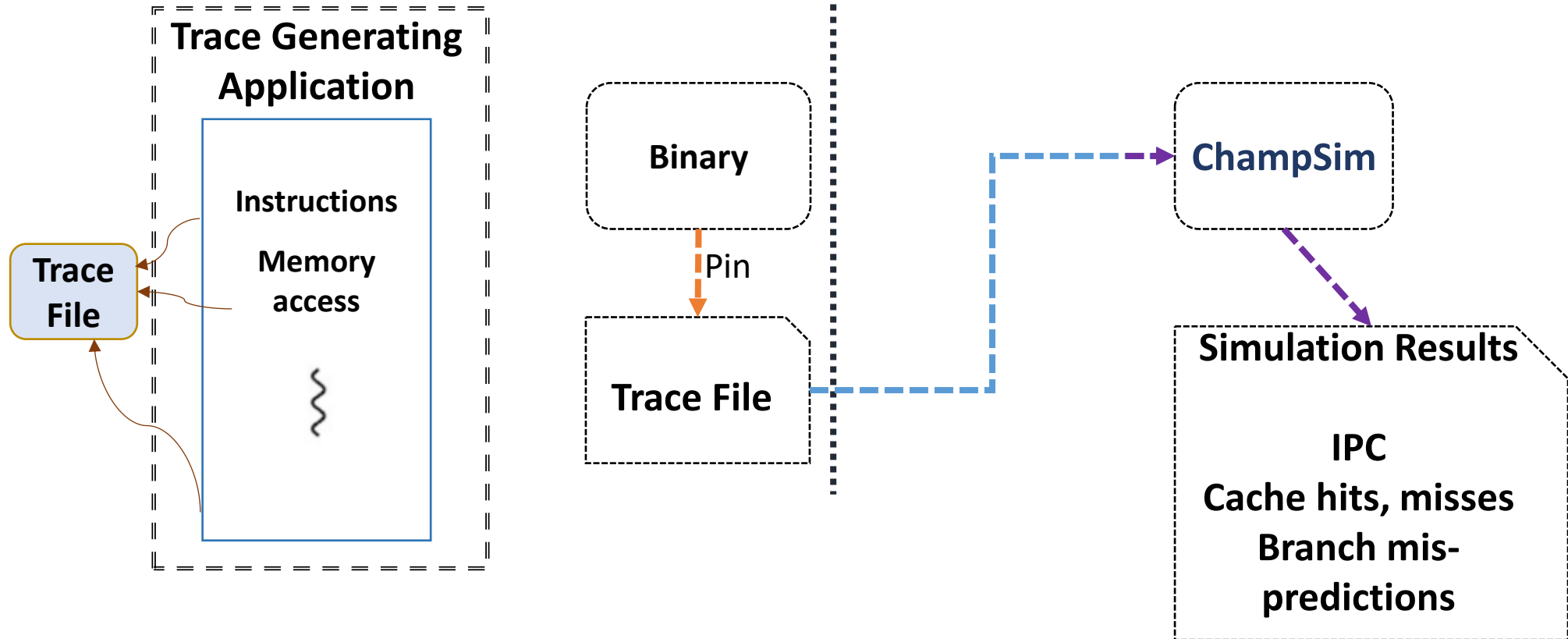
- Functional Simulator: Used to **verify the correct** execution of the program. Can not be used for performance evaluation.
- Performance simulators:
 - (i) Trace-driven: ChampSim
(<https://github.com/ChampSim/ChampSim>)
 - (ii) Execution-driven: gem5, Sinper, Multi2sim

Functional simulator is part of the performance simulators.

Intel Pin

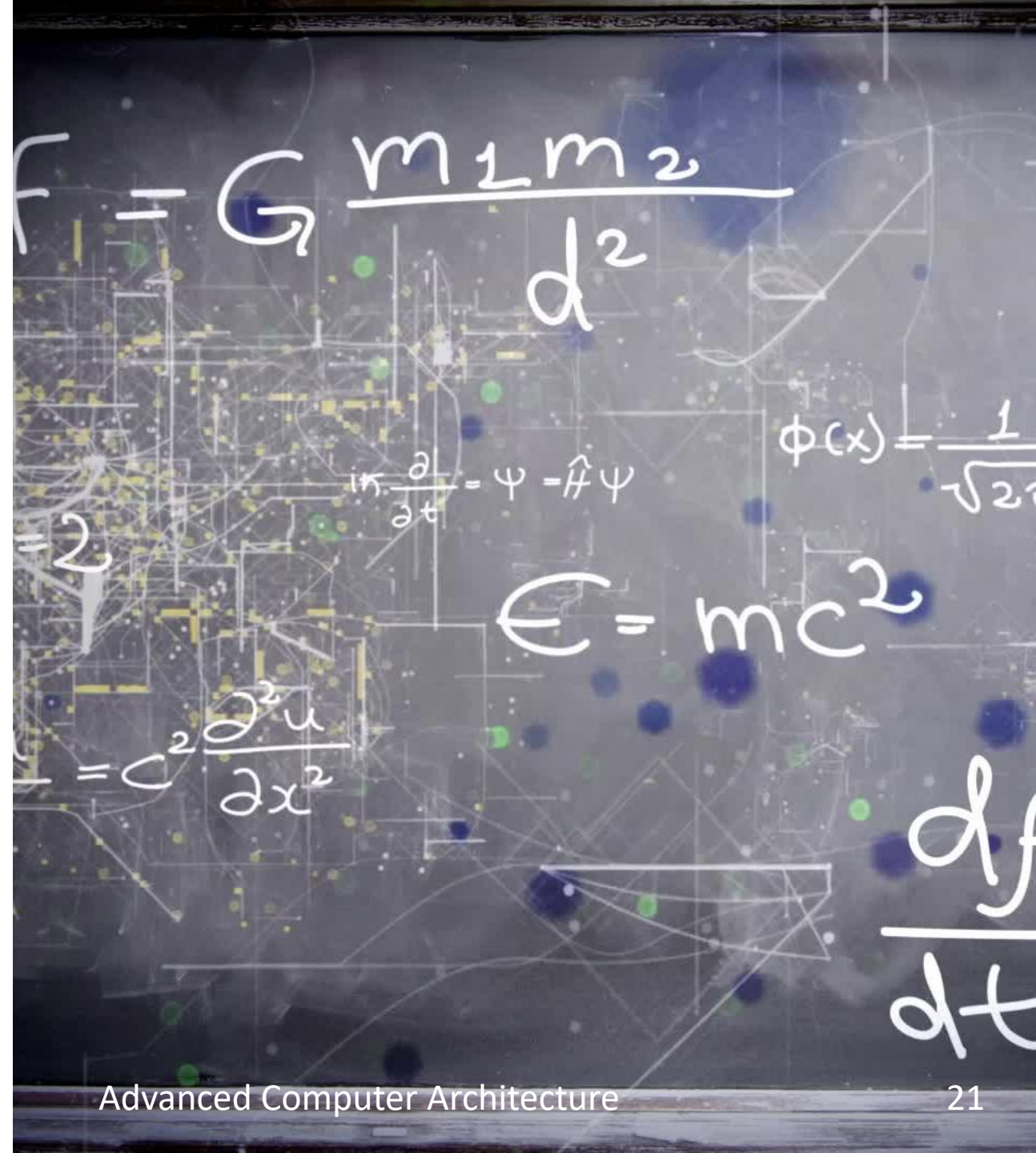
<https://www.intel.com/content/www/us/en/developer/articles/tool/pin-a-dynamic-binary-instrumentation-tool.html>

Trace based Simulator



Let's Dig Deeper into benchmarks

- Benchmarks: Billions of instructions
- Simulator is a software that simulates a hardware
- Speed of the simulator: 10M in five minutes say
- 100 billions: Months ☹ ☹





The idea of region of interest and sim-point

<https://cseweb.ucsd.edu/~calder/simpoint/>

Neat idea that saves time 😊

Let's discuss in detail

The notion of warmup and the academic crime ☹️

- Warmup microarchitecture structures before simulating and quantifying performance
- Tons of papers at low-quality venues: With no insights and ... ☹️ ☹️

Let's discuss in detail

Academic Crime ++

Use parameters that are old

Example, 256KB L3 cache when industry uses 2 to 3MB/core

Then, perform experiments, get some random insights, some adhoc conference, paper published, utility: Negative, not even zero

More academic crimes

- We did it because they did it and they did it because they did it 😞

Evaluation Continued

Pick a *relevant* benchmark suite

Measure IPC of each program

Summarize the performance using:

Arithmetic Mean (AM)

Geometric Mean (GM)

Which one to choose?

Example (Higher the better)

	IMTEL	ABM	AND
App. one	10	20	30
App. two	20	30	40
App. three	30	40	10

Which machine performs better over IMTEL and why?

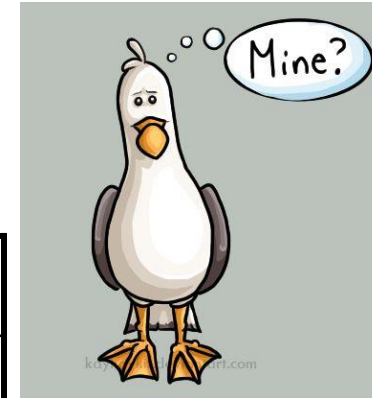
Contd.

	ABM	AND
App. one	2	3
App. two	1.5	2
App. three	1.3	0.3
A.M.	1.60	1.76
G.M.	1.57	1.21
H.M.	1.54	0.72

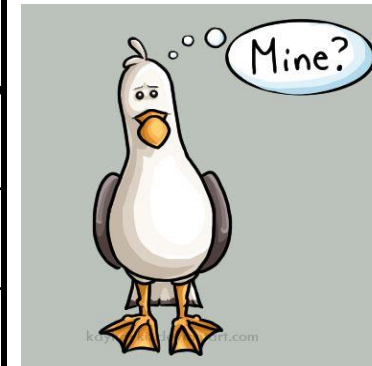


AM on ratios (use always GM on ratios)

	X	Y
App. 1	1	100
App. 2	1000	10
Normalized to X	X	Y
App. 1	1	100
App. 2	1	0.01
AM	1	50.005
Normalized to Y	X	Y
App. 1	0.01	1
App. 2	100	1
AM	50.005	1



Y is 50 times faster than X



X is 50 times faster than Y

Real System Experiments

- Noise
- Make sure the system state is kinda similar if not the same.
- Take the average of k experiments. For example: 5 to 10

Faithful Experiments

SUMS UP MY LIFE

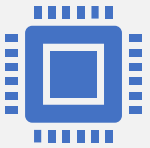
It doesn't work...
But why?



It works!
But why?



Some Mantra Again



MAKE SURE YOU CAN
DEFEND YOUR
RESULTS/CODE



ASSERT STATEMENTS 😊



WHY? WHY? AND WHY?



LET'S DISCUSS IN DETAIL

Performance@Multicore Systems

Application i running on an N -core system

Throughput = $\sum \text{IPC}(i)$

Individual Slowdown $(i) = \text{CPI-together}(i) / \text{CPI-alone}(i)$

Weighted Speedup = $\sum (\text{IPC-together}(i) / \text{IPC-alone}(i))$

Harmonic Mean of Speedups = $N / \sum (\text{IPC-alone}(i) / \text{IPC-together}(i))$

Unfairness =

Max-Slowdown/Min-Slowdown =

$\max(\text{Individual slowdowns}) / \min(\text{individual slowdowns})$

Start thinking about your project topics

- Browse recent flagship conferences
- Take a paper of interest
- Read it, do not know how to read a research paper

https://www.youtube.com/watch?v=LqJME-IUI_0

Decide a small task, Talk to TAs, me, CASPER group members

Submit your project proposal by October 1, 5 PM.